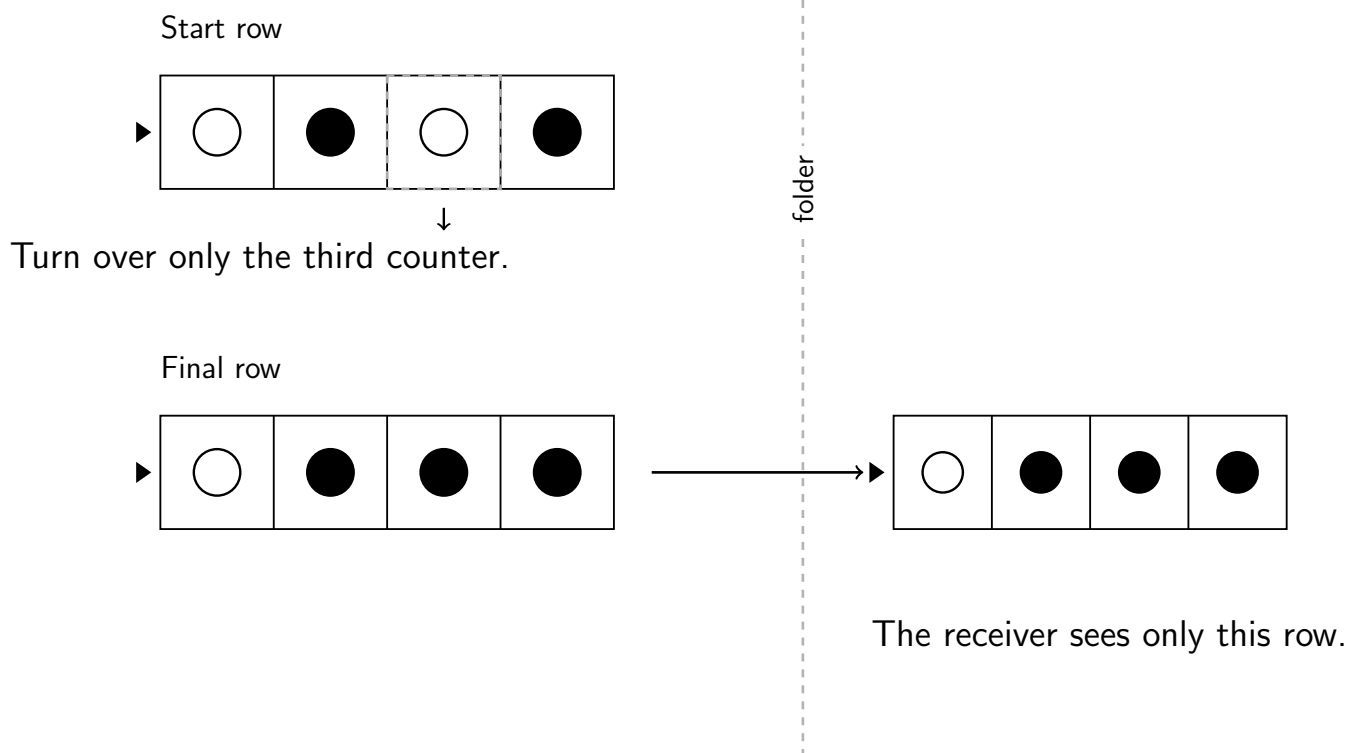


Keep the arrow at the left. The sender hides a picture and its row. The changer turns over zero or one counter in secret. Only the final row crosses the folder; the receiver uses the key to find the picture.

Example: a practice row

Sender and changer

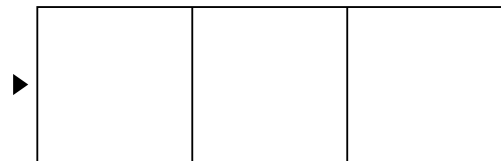
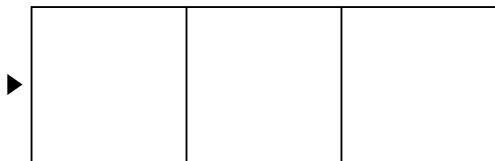
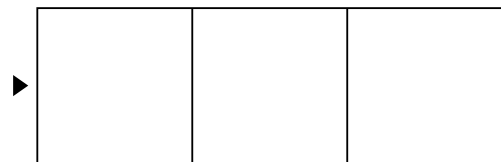
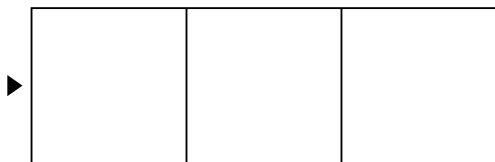
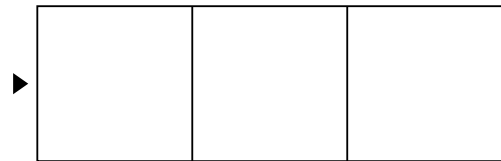
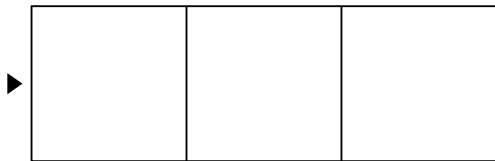
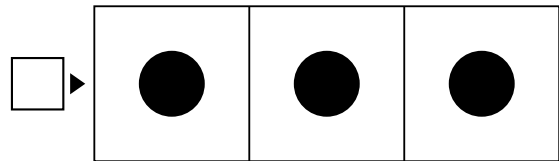
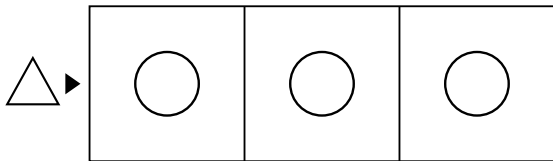
Receiver



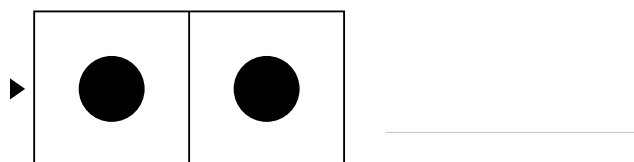
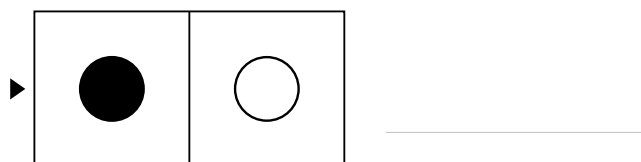
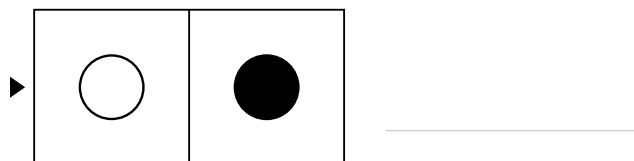
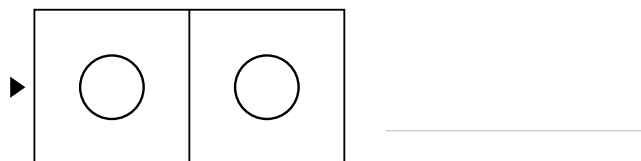
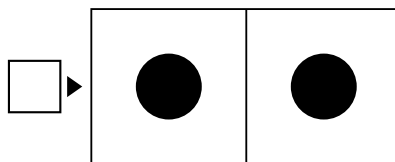
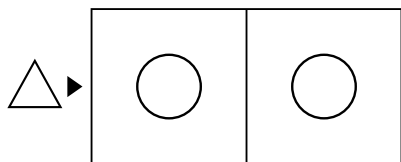
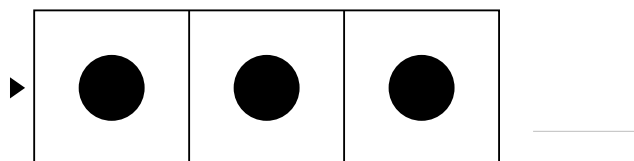
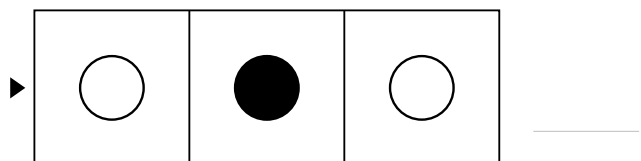
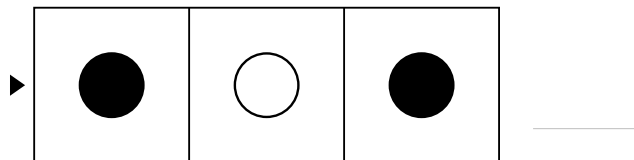
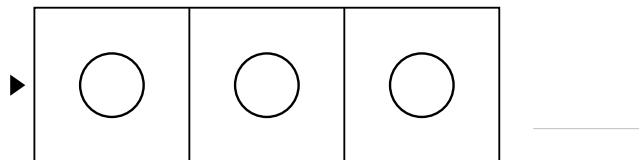
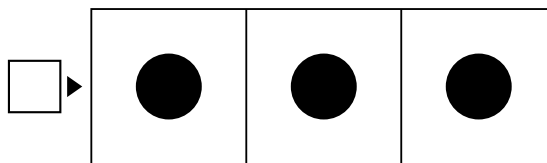
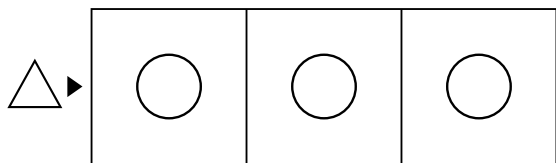
In the game, the start row and the changed position stay hidden behind the folder.

The changer may also leave the row as it is.

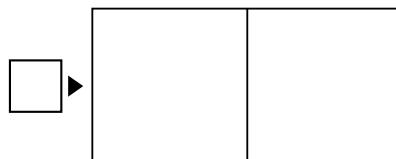
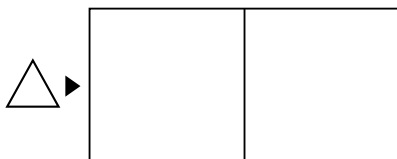
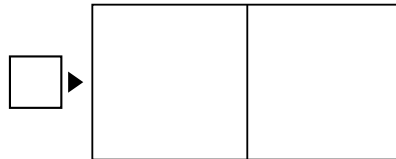
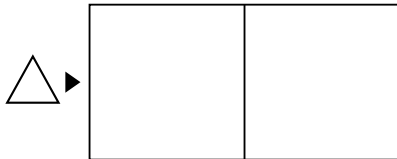
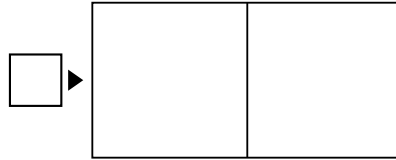
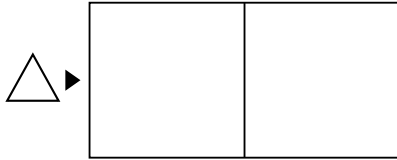
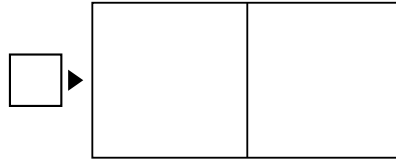
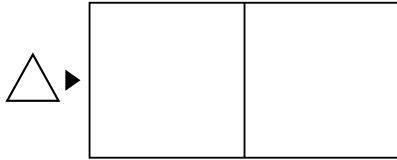
Problem 1: Send both pictures with this key many times. Can the changer fool the receiver?



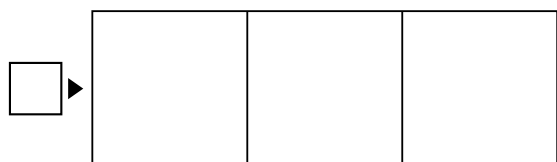
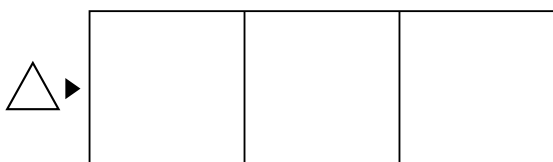
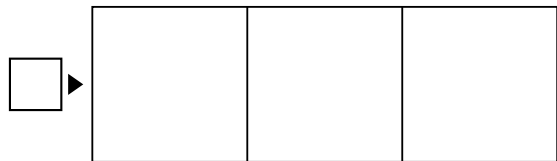
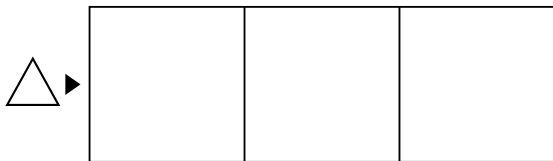
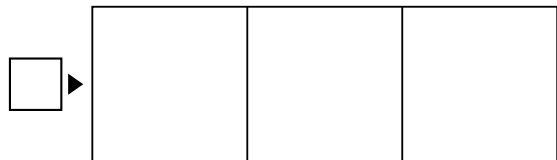
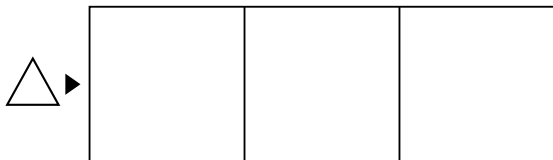
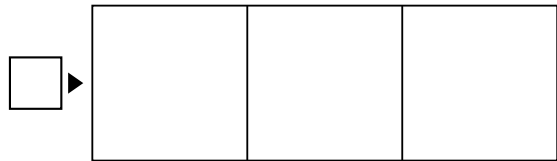
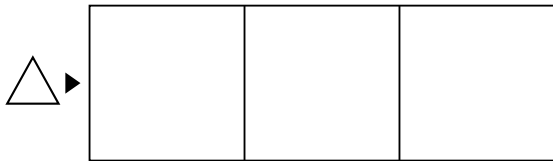
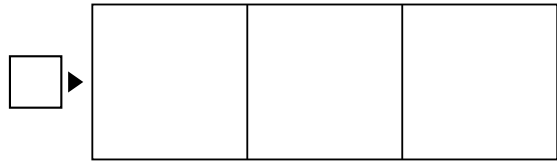
Problem 2: Draw every possible picture beside each final row. Which key lets the changer fool the receiver?



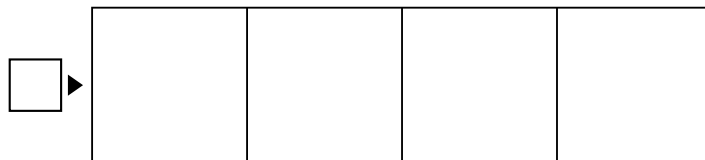
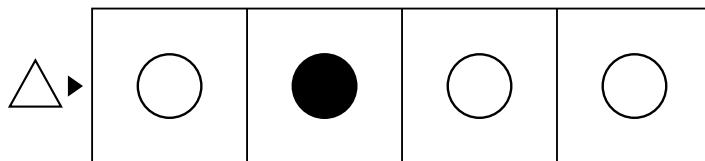
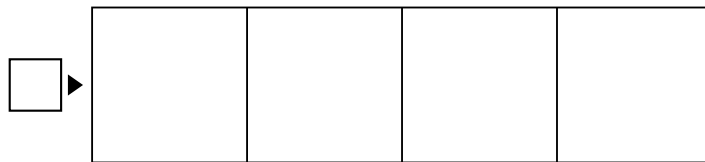
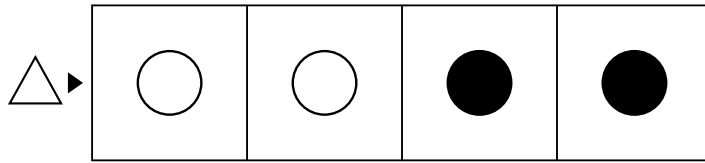
Problem 3: Make two two-counter rows for the pictures. Can you keep the changer from fooling the receiver?



Problem 4: Find every three-counter key that always tells the two pictures apart; use more paper if needed. Swapping the pictures makes a different key.



Problem 5: For each key, find every square row that always works. Use both kinds of counter in each row, and use more paper if needed.



Problem 6: Can three pictures share a three-counter key that always works?
Make one, or show why it cannot be done.

